

ORANGE, MA, USA

WMO: 725085

Lat: 42.570N

Lon: 72.291W

Elev: 555

StdP: 14.40

Time Zone: -5.00 (NAE)

Period: 96-19

WBAN: 54756

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.7	2.7	-13.4	2.7	1.6	-8.6	3.5	6.0	23.3	23.9	20.6	24.3	2.7	290	0.519

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.9	89.6	71.6	86.4	70.0	83.6	68.6	74.7	83.5	73.2	81.2	71.8	79.3	9.6	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.2	121.7	78.3	70.9	116.1	76.9	69.7	111.5	76.0	38.4	83.6	37.1	81.8	35.9	78.7	80.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.9	17.6	15.8	DB	-10.9	94.3	6.6	3.0	-15.7	96.5	-19.5	98.3	-23.2	99.9	-28.1	102.1	(4)
(5)			WB	-11.3	77.3	6.5	1.3	-16.0	78.2	-19.8	79.0	-23.5	79.7	-28.2	80.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.5	22.7	25.5	33.6	45.5	56.9	65.4	70.9	69.2	61.7	49.3	38.5	29.3		
	DBStd	18.56	10.87	9.66	9.22	8.30	7.75	6.75	5.48	5.49	7.22	8.05	8.49	9.01		
	HDD50	3367	847	685	515	184	20	0	0	0	3	114	357	642		
	HDD65	6899	1312	1105	974	586	274	77	12	22	147	490	794	1106		
	CDD50	2454	0	0	6	50	234	462	648	595	354	91	12	2		
	CDD65	509	0	0	0	2	22	89	194	152	48	2	0	0		
	CDH74	5040	0	0	7	68	379	921	1906	1333	405	21	0	0		
	CDH80	1557	0	0	1	21	113	293	657	370	100	2	0	0		
(14) Wind	WSAvg	5.5	5.8	6.3	7.0	6.7	5.9	5.2	4.8	4.3	4.3	5.1	5.4	5.6		
(15) Precipitation	PrecAvg	47.2	3.6	3.1	3.9	3.7	3.8	4.7	3.9	4.1	4.2	4.3	3.8	4.2		
	PrecMax	62.6	6.0	10.5	6.8	6.5	5.9	9.3	8.0	9.0	11.3	14.1	5.9	7.9		
	PrecMin	34.6	1.4	1.4	0.9	1.0	1.4	0.7	1.5	1.4	0.6	0.8	1.3	1.7		
	PrecStd	8.3	1.1	1.9	1.7	1.7	1.2	2.2	1.9	1.8	2.6	2.8	1.1	1.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.6	57.1	70.9	82.5	88.5	91.4	93.6	91.1	88.2	76.7	66.1	60.9		
		MCWB	52.1	49.6	56.1	61.1	66.0	71.6	74.7	73.2	71.8	63.6	59.1	55.7		
	2%	DB	46.5	49.5	59.6	73.0	82.9	87.6	89.8	87.4	82.4	72.0	62.3	52.5		
		MCWB	42.3	42.6	48.4	55.0	64.6	70.1	72.1	72.0	68.6	62.6	56.8	49.2		
	5%	DB	41.2	44.1	53.4	66.9	78.3	82.9	86.6	84.0	78.7	67.8	57.6	47.5		
		MCWB	37.2	38.1	44.4	53.0	61.4	67.4	70.6	69.7	66.6	60.7	52.4	44.0		
	10%	DB	36.9	38.8	47.7	62.5	72.8	79.5	83.6	81.3	74.9	63.4	53.7	42.9		
		MCWB	33.6	34.1	41.3	50.0	59.0	65.1	69.2	68.3	65.1	56.6	48.3	38.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.3	53.6	58.2	64.1	70.2	75.0	77.2	76.4	74.1	69.0	62.3	55.7		
		MCDB	54.2	55.8	68.1	74.8	82.5	85.0	89.5	85.9	81.4	73.6	64.6	57.7		
	2%	WB	43.5	43.9	51.3	59.8	67.3	72.8	74.9	74.4	71.8	65.2	57.9	50.1		
		MCDB	45.5	48.3	57.1	69.6	77.0	82.2	84.4	82.2	78.2	69.0	60.9	52.5		
	5%	WB	37.7	38.3	45.9	55.7	64.8	71.1	73.2	72.9	70.0	61.8	53.6	44.3		
		MCDB	40.4	42.8	51.3	62.8	73.5	79.1	81.6	80.0	75.5	66.4	56.8	46.7		
	10%	WB	34.1	35.1	41.6	51.7	62.0	68.7	71.8	71.2	67.6	58.1	48.8	39.2		
		MCDB	36.3	38.1	47.6	59.7	69.7	75.3	79.6	77.7	73.0	62.6	52.9	42.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	18.0	19.8	19.3	23.1	23.2	21.5	21.9	21.6	22.2	20.9	18.8	16.6		
		MCDBR	20.0	24.4	29.1	33.4	32.5	27.4	25.3	24.4	25.6	25.5	24.0	21.6		
	5% WB	MCWBR	17.0	18.5	18.4	17.4	15.2	12.1	10.0	10.6	13.1	15.8	18.2	18.8		
		MCWBR	19.1	22.1	25.4	27.3	25.9	22.6	21.5	19.3	20.0	21.2	20.9	20.2		
(40) Clear-Sky Solar Irradiance	taub		0.284	0.294	0.321	0.366	0.399	0.442	0.460	0.429	0.374	0.332	0.303	0.283		
	taud		2.442	2.416	2.442	2.372	2.316	2.221	2.179	2.252	2.387	2.520	2.555	2.534		
	Ebn at Noon		274	290	294	286	278	265	259	263	272	272	266	266		
	Edh at Noon		24	29	32	37	40	44	46	41	33	26	22	20		
(44) All-Sky Solar Radiation	RadAvg		553	823	1157	1462	1615	1746	1840	1652	1344	863	600	448		
	RadStd		40	70	79	157	186	168	114	85	96	88	53	47		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+243	+103	
(47) Regional (76 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+169	+76	

Nomenclature: See separate page